

CROSS SELLING LEAD

Building a Scalable Customer Data Pipeline for Cross Selling

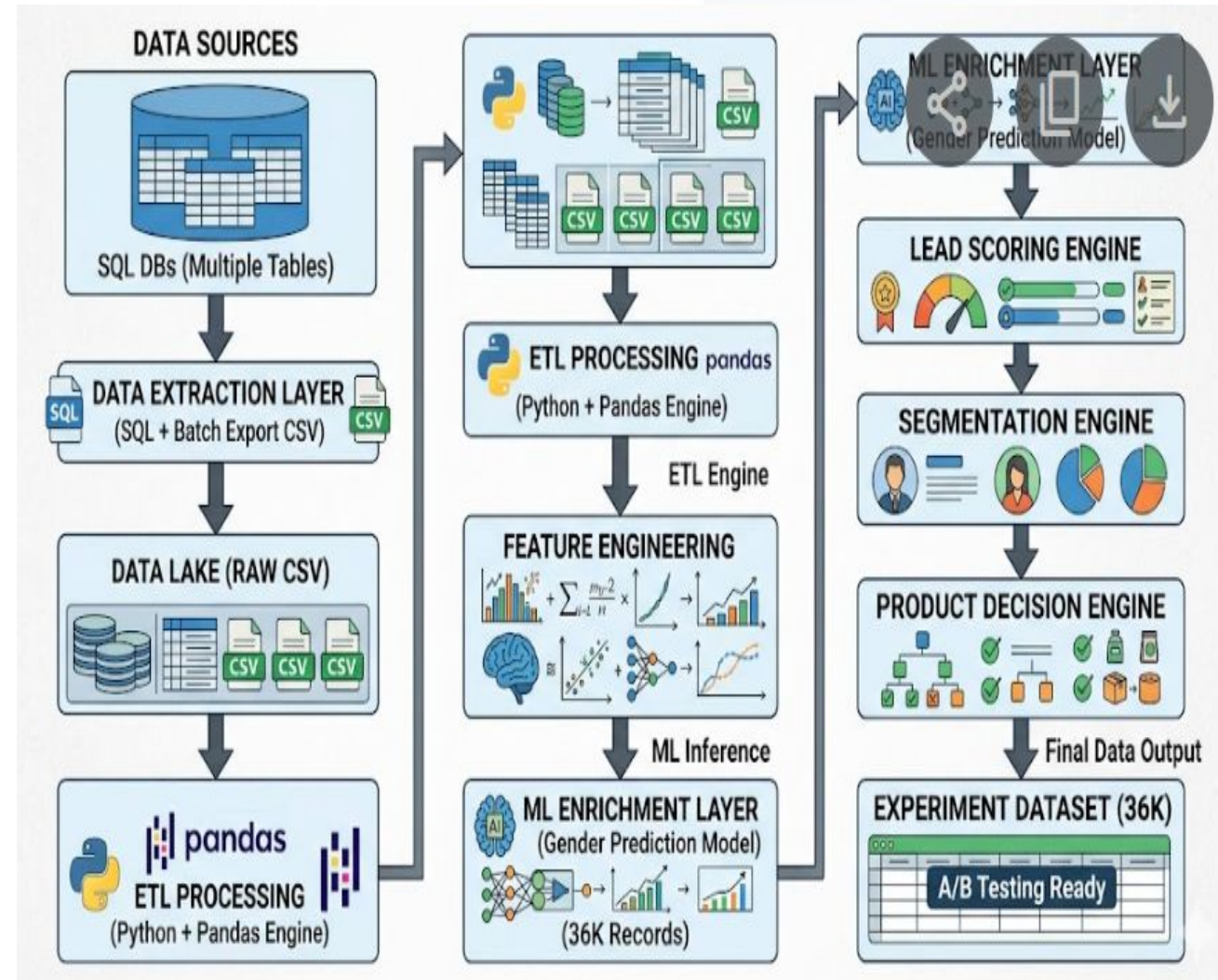
Consolidate | Predict | Segment | Allocate

Problem Context: Data Fragmentation

The Challenge

Organizations often hold millions of customer records trapped across fragmented databases. This lack of a "Unified Customer View" prevents intelligent cross-selling.

Key pain points include redundant records, inconsistent attributes, and the absence of demographic signals required for personalization.



The Solution: Intelligent Pipeline



Consolidate

Merge multi-source customer data into unique identities using mobile key aggregation.



Enrich

Predict missing attributes like gender and classify into behavioral segments.



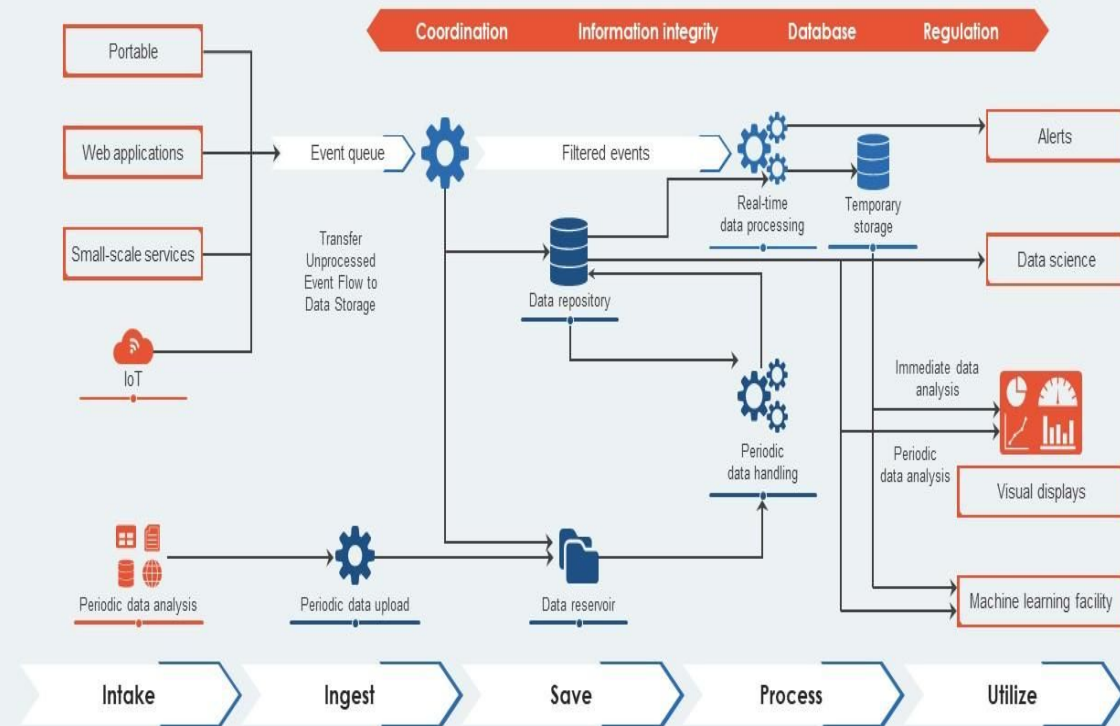
Allocate

Assign optimal products based on lead quality scores and city tiers.

End-to-End System Architecture

Big data pipeline architecture for efficient process

The purpose of this slide is to illustrate how Big Data pipeline architecture for efficient processing optimizes data workflows, facilitating faster analytics and insights generation. It includes batch data, load, data lake, etc.



This slide is 100% editable. Adapt it to your needs and capture your audience's attention.

Scalable Modular Design

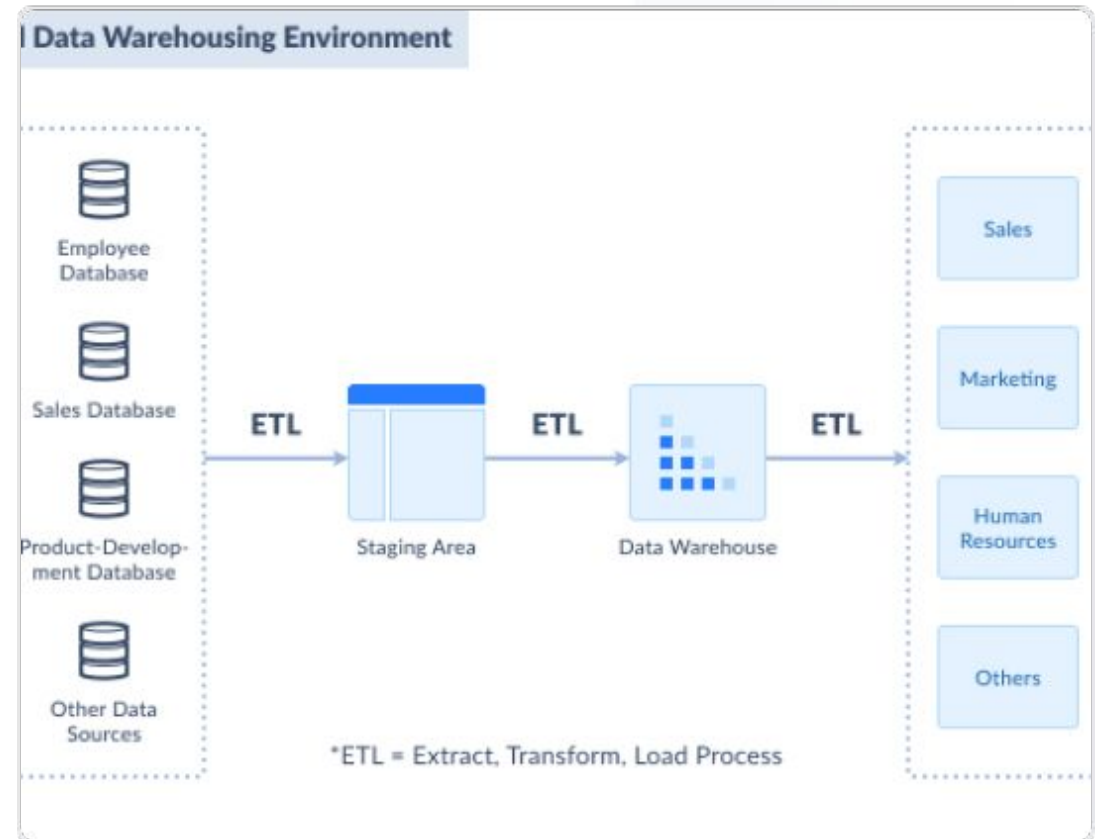
The pipeline is designed to ingest 7.5 million records through an ETL layer, applying NLP and Machine Learning models to derive actionable outreach lists. This architecture decouples raw data extraction from logical decision-making, ensuring maintainability.

Phase 1: Efficient Data Extraction

Relational to Analytical

Extracting from Multiple SQL Databases using batch-wise processing to respect memory constraints.

- SQL Joins + UNIX Timestamp filters.
- 7.5 Million total raw records.
- Year-wise batches to optimize RAM usage.



Process Pipeline (Steps 1-6)



The initial phase focuses on converting unstructured noise into a clean, feature-rich relational structure for Machine Learning.

Process Pipeline (Steps 7-12)



The intelligence phase transforms raw leads into prioritized outreach segments using specialized predictive models.

Ingest, Aggregate & Flatten



Deduplication Logic

We group data by **Mobile Number** to ensure one row per customer. This is the bedrock of our pipeline.

- Aggregates car interests via lambda join.
- Flattens multi-level columns.
- Retains "Max Price" as buying power proxy.

Phase 2

Gender Prediction Model

Natural Language Processing for Demographic Inference

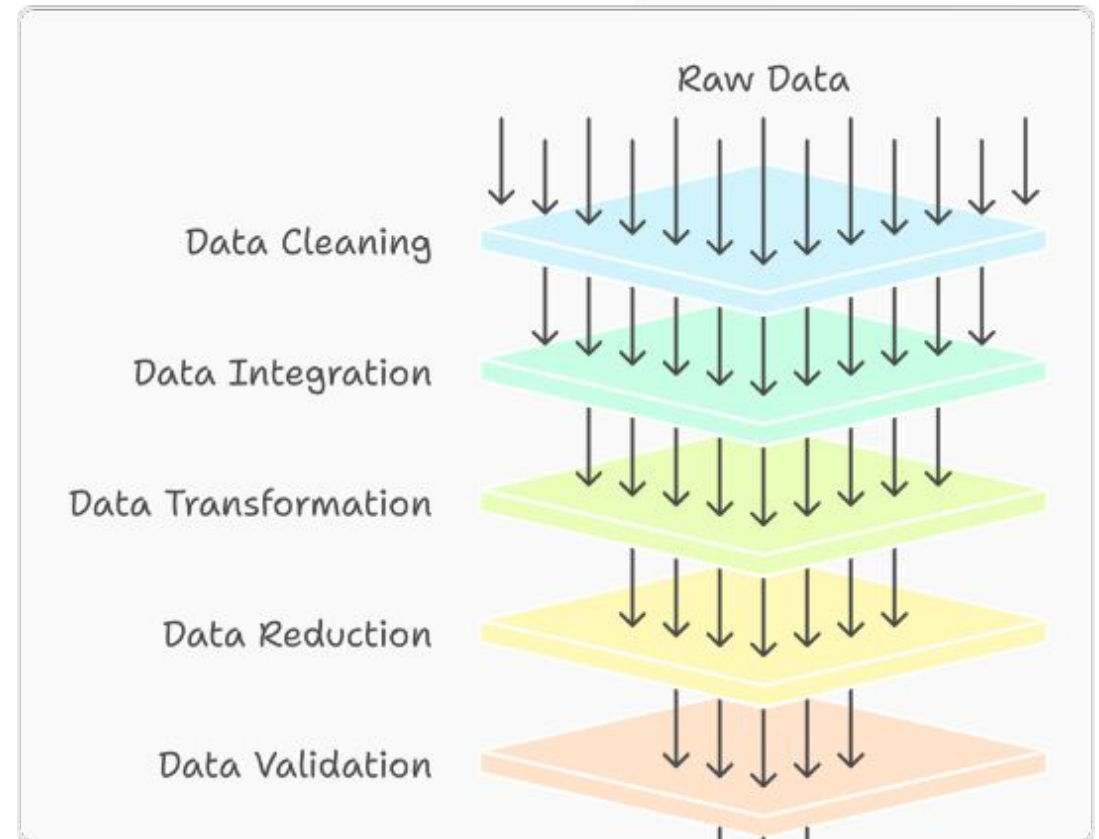


Gender Model: Text Cleaning

Preparing Names for Vectorization

Standardizing names to remove noise and resolve gender conflicts between conflicting datasets.

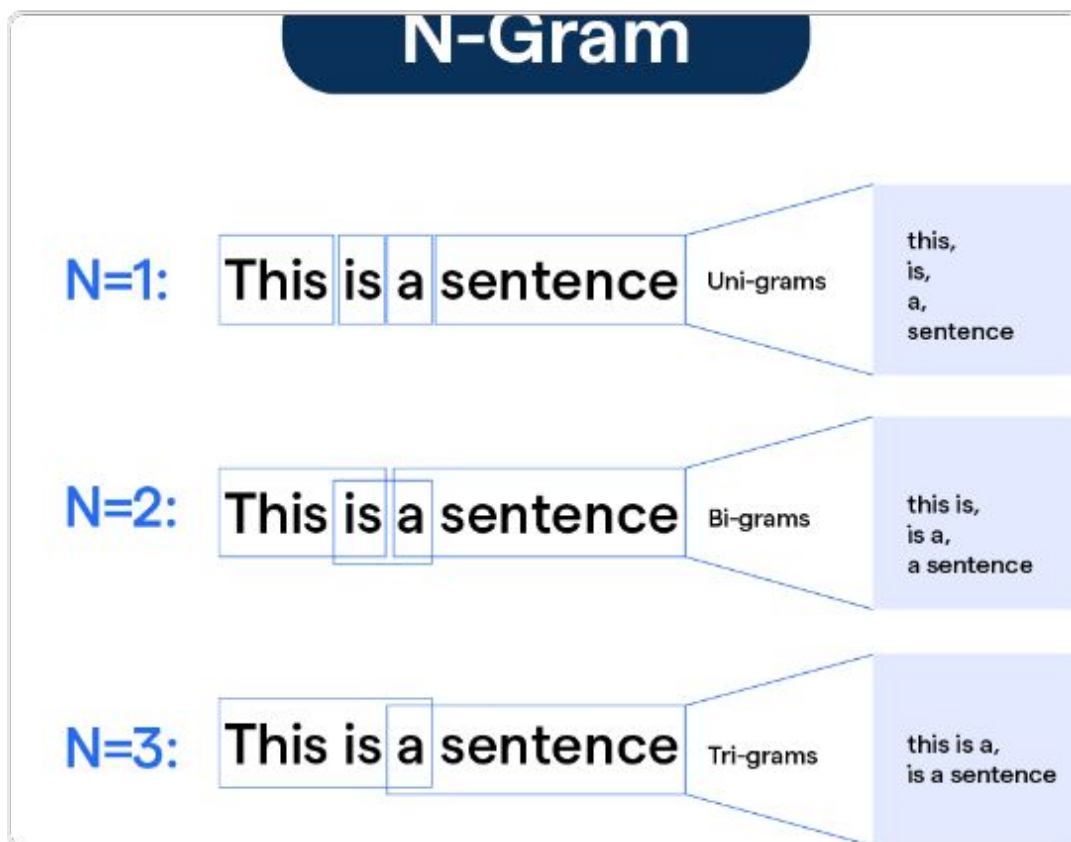
-  Lowercasing & Symbol removal.
-  First Name Extraction via Regex.
-  Conflict Resolution: Male > Female bias.



Gender Model: Feature Engineering

The Power of Character N-Grams

Philippine names follow distinct morphological patterns. We use **Character N-grams (2-5)** to capture gender signals.



Name: MARIA

2-grams: ma, ar, ri, ia

3-grams: mar, ari, ria

Gender Model: Vectorization

TF-IDF: Term Importance

Converts character patterns into numerical matrix by weighing their frequency against commonality.

$$\text{TF-IDF} = \text{TF} \times \log \frac{N}{\text{df}}$$

This allows the model to ignore common "noise" patterns and focus on high-signal gender markers.

TF-IDF

TF-IDF is a measure of originality of a word by comparing the number of times a word appears in a doc with the number of docs the word appears in.

$$\text{TF-IDF} = \text{TF}(t, d) \times \text{IDF}(t)$$

Term frequency

Number of times term t appears in a doc, d

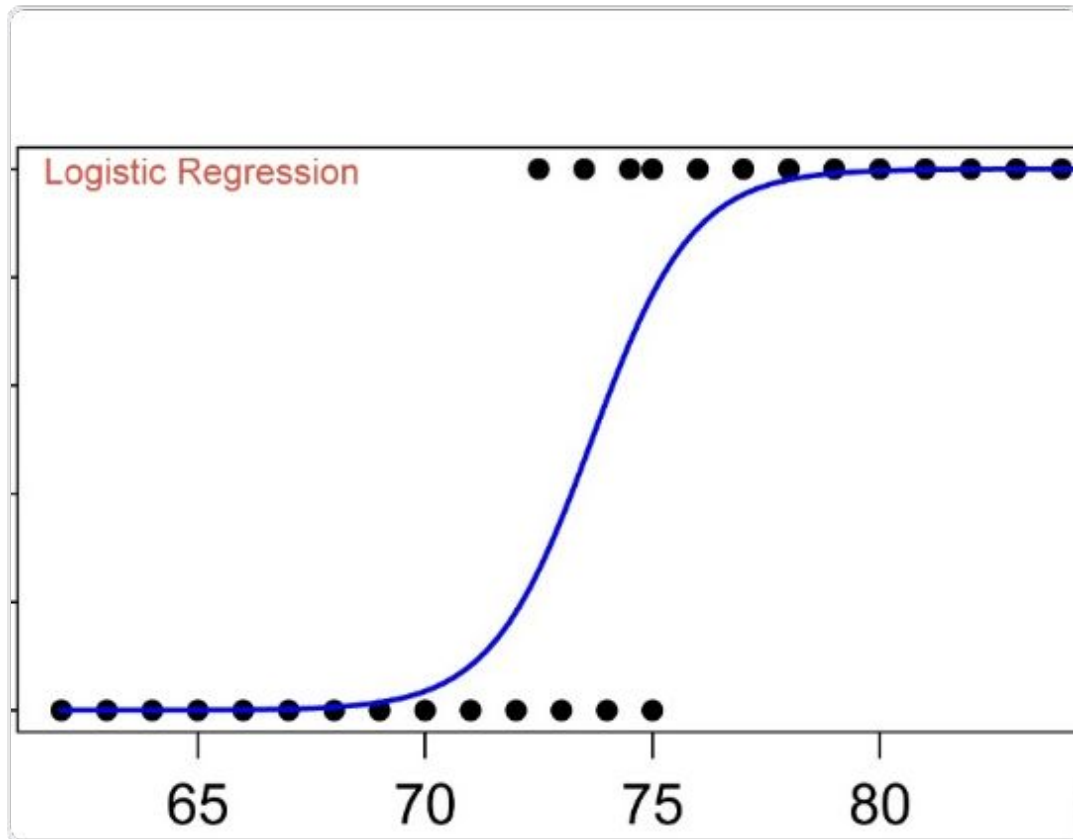
Inverse document frequency

$\log \frac{1 + n}{1 + \text{df}(d, t)}$

n ← # of documents

Document frequency of the term t

Classifier: Logistic Regression



Probability Estimation

Logistic Regression acts as a linear classifier predicting the probability of a label using the Sigmoid function.

$$p = \frac{1}{1 + e^{-z}}$$

Resulting Output: 0 for Female, 1 for Male.

Evaluation: Accuracy & CV



96.8% Accuracy

Model is highly reliable on unseen Philippine names.

Zero Overfitting

CV score consistent with test accuracy.

Performance: Confusion Matrix

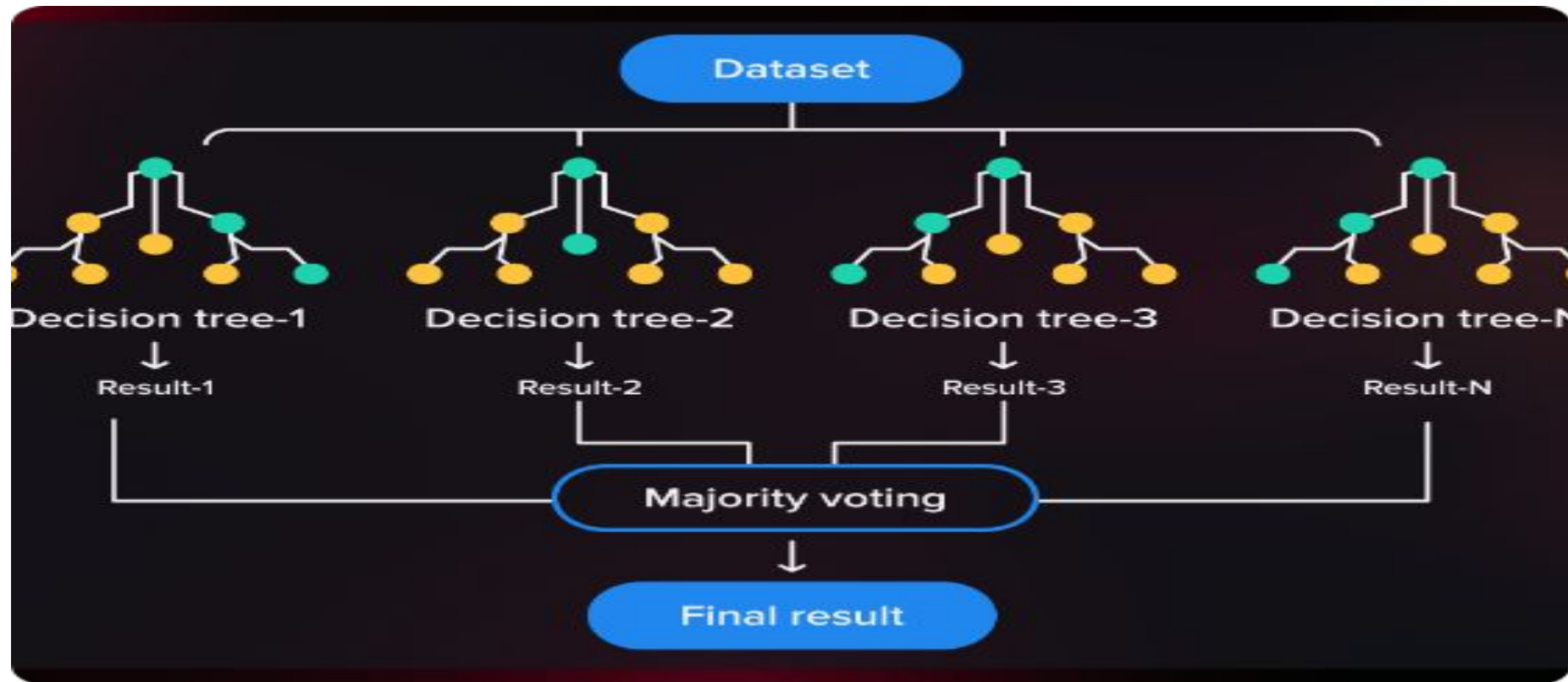
N=817		
	Predicted Male	Predicted Female
Actual Male	404	8
Actual Female	15	390



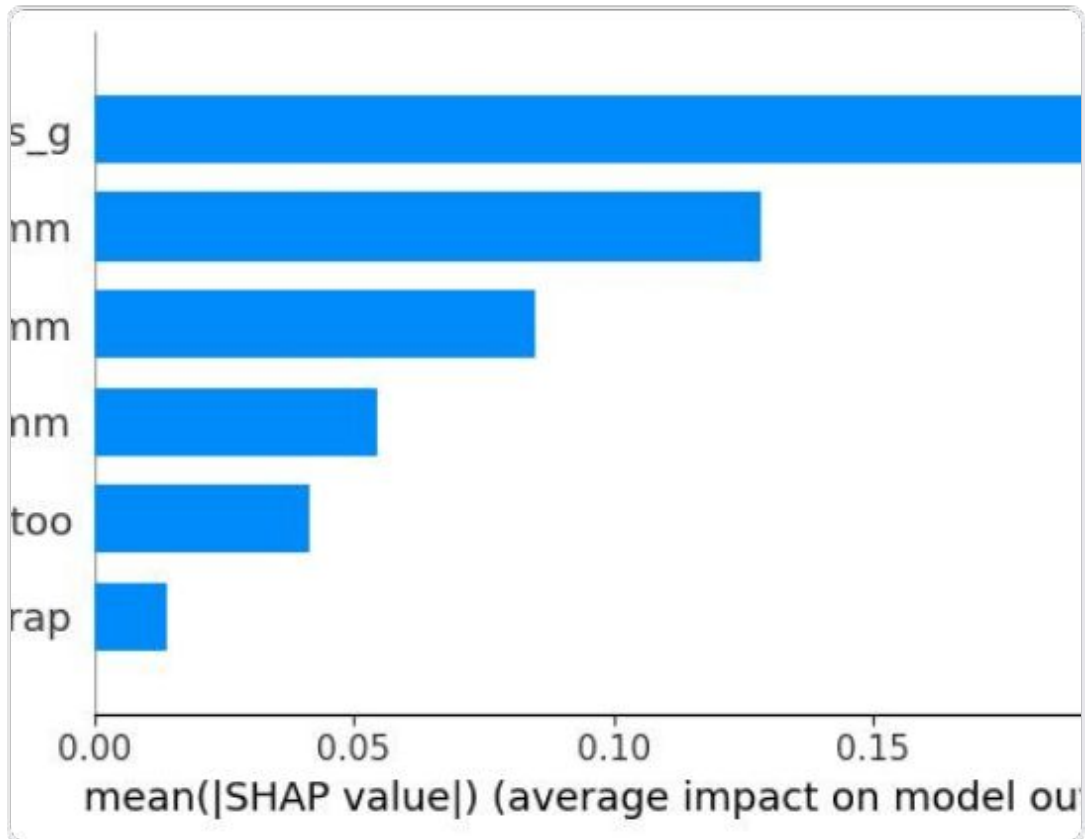
Phase 3

Lead Quality Scoring

Random Forest Regression for Value Prediction



Scoring: Feature Importance



Deriving Objective Weights

We use a **Random Forest Regressor** to determine which attributes most accurately predict purchasing power.

- Max Price (Car Value)
- Inquiry Volume (Engagement)
- City Tier (Economic Strength)

Prioritizing the Elite Tier

Q4

Elite Prospects

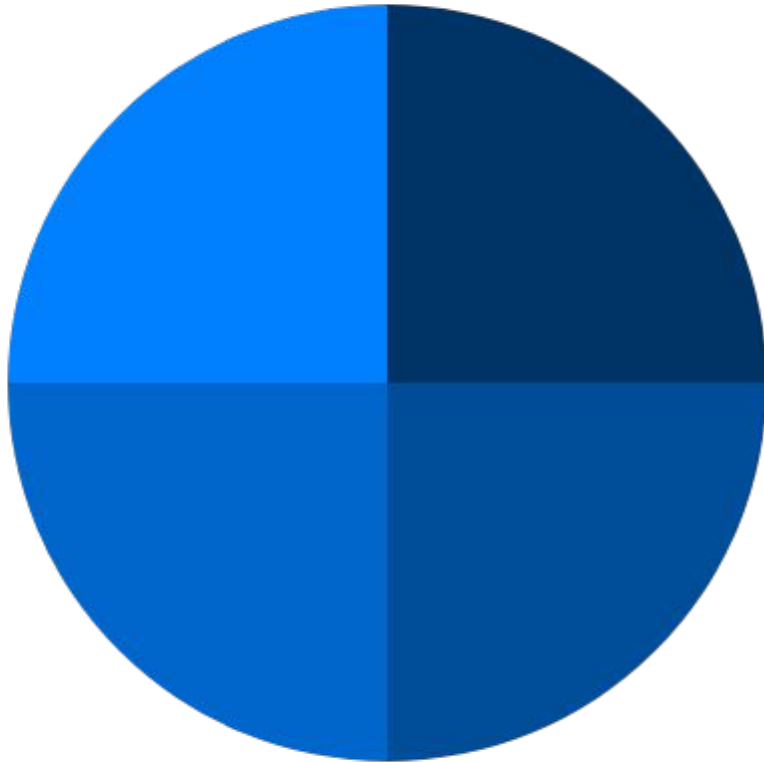
Segmentation via Quartiles

Lead scores are bucketed into four distinct quartiles.

We prioritize **Q4 (Elite)** and **Q3 (Strong)** for high-value product cross-selling.

This allows the sales team to focus efforts on the top 25% of the lead pool with the highest conversion probability.

Customer Segment Composition



Balanced Segments

The lead pool is divided equally to test conversion across different economic profiles.

- ◆ Elite Outliers (Top Tier)
- ★ High-Potential Leads
- 📈 Growth Prospects

Phase 4

Product Allocation Engine

Deterministic Rule Matrix for Market Fit

Product-Segment Alignment

Customer Segment	City Tier 1/2	City Tier 3
High-End Outlier	Product 1	Product 1/Product 2
Elite Prospect	Product 1/Product 2	Product 2
Strong Prospect	Product 2	Product 2/Product 3
Growth Prospect	Product 2/Product 3	Product 3

Phase 5: Stratified Sampling



Testing for incorrect duration



Using the wrong tool



Failing to follow an iterative process



Failing to consider external factors

Ensuring Statistical Rigor

Final output is a perfectly balanced **36,000 Dataset** for unbiased A/B testing.

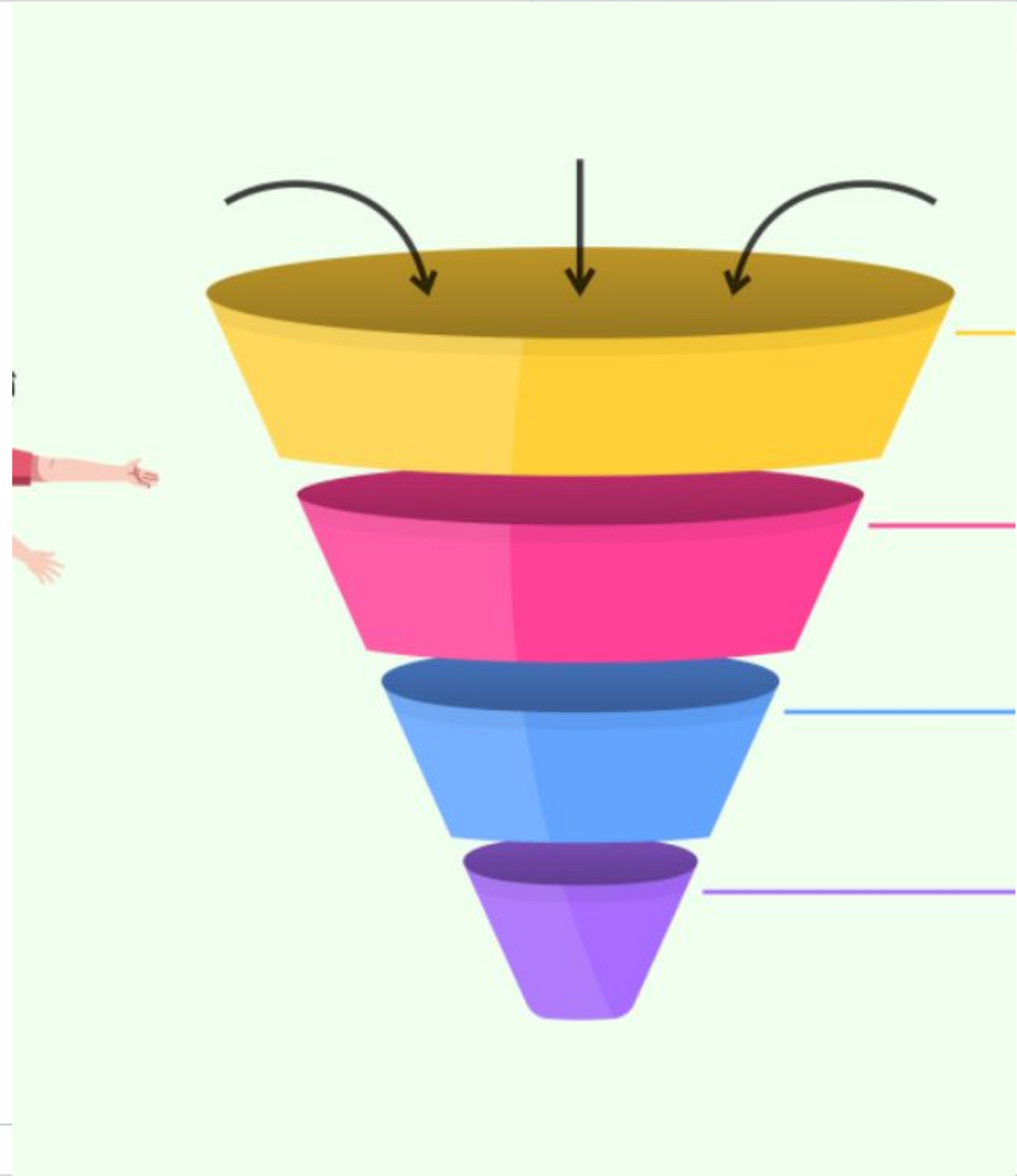
- 12,000 per product category.
- Strict 50/50 Gender split per bucket.
- Zero overlap across groups via unique UID set.

Export & Outreach Ready

The Final Sample: 36K

This dataset is shuffled and exported for direct integration with the marketing outreach tools.

By controlling for gender and product proportions, the business can accurately measure the uplift provided by the ML-driven lead scoring system compared to traditional random selection.



Key Machine Learning Concepts



Evaluation

Precision/Recall: Balancing capturing all leads vs ensuring they are the right ones.



Validation

K-Fold CV: Ensuring model performance is consistent across different data splits.



Sampling

Stratification: Maintaining the same distribution of classes in train and test sets.

Questions?

Transforming Fragmented Data into Cross-Selling Success



Key Performance Indicators:

- 96.8% Model Accuracy
- 7.5M Record Pipeline Scalability
- Zero-Overlap A/B Test Design